CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	[Destiny Johnson Novemeber 9, 2026]
Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

R03 is marked READY but has no Inspector_ID, which fails Q05. According to W07-S02, every READY record must identify an inspector. This missing information prevents users from identifying who completed the inspection and requires verification.

Next action: what to verify, who owns it, and how to retest

The Shift Supervisor should verify R03's READY status and inspector assignment against the original inspection record. Once a correction is confirmed, I would refresh the query and retest Q05 to check that a READY record includes an Inspector_ID.

Limit: why you cannot guess a correction or prove accuracy

I cannot guess the missing Inspector_ID because the spreadsheet does not establish who performed the inspection. Guessing could create a false record. Passing these rules only confirms that the recorded values meet the requirements; it does not prove the inspection occurred or that the machine is safe.

Professional connection: one framework idea and how you used it

The Government Data Quality Framework recommends using clear rules and measurable results to assess data quality. I applied this idea by checking all 20 records against Q01–Q05 and reporting affected rows, counts, and percentages. This makes the issues clear and helps guide follow-up.

Refresh test: before, changed value, after, and restored result

In a temporary copy, Q04 initially flagged 3 rows (15%). I changed R01's Inspection_Minutes from 20 to 0 and selected Refresh All. Q04 increased to 4 rows (20%). After restoring the value to 20 and refreshing again, Q04 returned to 3 rows (15%). This confirmed that the query and audit results updated when the source data changed.

Export only Report: two landscape pages, page break before row 26. Review both pages.